Formulating Research Topic

What is Research

- Research is a **systematic** and **methodical** process of inquiry aimed at *exploring*, *analyzing*, and *interpreting* information to answer specific questions or solve problems.
- Research has many definitions:
  - *Research is a careful, systematic and objective investigation conducted to obtain valid facts, draw conclusions and establish principles regarding an identifiable problem in some field of knowledge.*

Characteristic of Research

- **Systematic Approach:** Research follows organized steps to ensure logical data collection, analysis, and interpretation.
- **Objective and Unbiased:** Conclusions are drawn based on evidence rather than personal opinions.
- **Empirical Evidence:** Observations, experiments, and surveys provide the foundation for findings.
- **Clear Research Question:** Every study begins with a specific problem or question guiding the process.
- **Replicability:** Others should be able to reproduce the study and achieve similar results.
- **Transparency and Ethics:** Researchers adhere to ethical standards, ensuring informed consent and confidentiality.

Characteristic of Research (cont.)

- **Generalizability:** Findings aim to apply beyond the studied sample or context.
- **Logical and Critical Thinking:** Data is analyzed carefully to identify patterns and insights.
- **Contribution to Knowledge:** Research advances understanding, challenges theories, or proposes new ideas.
- **Peer Review and Publication:** Findings are validated through expert review and shared in journals or conferences.
- **Iterative Process:** Research often leads to new questions and further studies.
- **Practical Application:** Results may inform policy, improve practices, or solve real-world problems.

Types of Research

- Quantitative Research
- Qualitative Research
- Applied Research
- Basic Research
- Correlational Research
- Descriptive Research
- Experimental Research
- Exploratory Research
- Ethnographic Research
- Mixed-methods Research

Quantitative Research

- Centered around collecting numerical data and employing statistical techniques to draw conclusions.
  - Experimental Research: Testing a new medicine by giving it to one group and placebo to another group.
  - Correlational Research: Tracking the number of hours students exercise and recording their test scores to see if a relationship exists.
  - Descriptive Research: Counting the average daily income of families living in a specific city.

Qualitative Research

- focuses on understanding the intricacies of human experiences, beliefs, and social phenomena.
  - Interviews: Long, open talks with people to learn about their personal views.
  - Focus Groups: Group chats led by a host to see how people talk and share ideas about a topic.
  - Observations: Watching people in their real, everyday spaces to see how they act.

Applied Research

- It is a scientific study that seek to solve various practical problems in the day to day life
  - Medical Image Analysis
  - Dynamic Pricing Algorithms
  - Supply Chain Demand Forecasting

Basic Research

- Also called as fundamental or pure research; not create or invent new things
  - Universe origin theory: How did universe begin
  - New machine learning optimization techniques

Correlational Research

- Study the relationship among 2 or more variables; can be causal relationship or a simple negative/positive relationship
  - Correlation between screen time and sleep
  - Correlation between music and brain reaction time
  - Correlation between solar farms and localized environmental changes

Descriptive Research

- provides a clear and accurate profile of a person, event, or group. Also called statistical research
  - A survey measuring how many hours college students spend studying during final exam week
  - A medical case report detailing the unique symptoms and progress of a single patient with a rare condition.
  - A census tracking the age, job status, and housing types of people living in a specific city.

Experimental Research

- Also called *Case-Control* research; compares individuals with a particular outcome (cases) to those without it (controls) to identify potential causes or risk factors
  - Drug Trials: Give one patient group a new blood pressure medicine and give the control group a fake sugar pill (placebo) to see if the real drug works.
  - Website Layout: Show half of web visitors a blue checkout button and the other half a red checkout button. Count which color gets more clicks.

Exploratory Research

- conducted for a problem that has not been clearly defined; helps to determine the best research design, data collection method and selection of subjects.
  - Customer Segmentation: Grouping unlabeled e-commerce buyer data using clustering tools to see what natural buyer personas emerge.
  - Anomaly Detection in Logs: Inspecting server traffic histograms and scatter plots to find weird spikes or system errors before defining a fixed alert rule

Ethnographic Research

- involves investigation of a culture through an in-depth study of members of culture
  - studies people, ethnic group, ethnic formations and social welfare characteristics.

Mixed-Methods Research

- combines qualitative and quantitative approaches to gain a deeper insight into complex problems.
  - Recommending Systems and User Experience (UX)
    - **Quantitative Phase:** Analyze petabytes of clickstream data and watch time to track user engagement trends with collaborative filtering models.
    - **Qualitative Phase:** Run user interviews to see how people feel about recommendations and why certain algorithms feel "creepy" or unhelpful.
    - **Integration:** Tune the quantitative recommendation engine to factor in human feedback on privacy and preference.

Formulate a Dissertation Topic

- Find Your Focus
  - Pick a broad field: Choose a subject or class theme you enjoy.
  - Review current literature: Read recent journal articles and check the "future work/direction/research" or “research limitation” sections.
  - Identify a gap: Look for missing data, new target groups, or untested theories, open questions, unsolved problems.
  - Use the PICK approach: Define your People (target group), Issue (core problem), and Context (setting or environment)

Formulate a Dissertation Topic

- Test for Feasibility
  - Data access: Make sure you can actually reach the data, people, or tools you need.
  - Time limits: Pick a scope you can finish in your program schedule.
  - Ethical approval: Check if your school will easily approve human or animal study rules
  - Interest level: Pick a topic that keeps you curious, but avoid overly emotional or personal subjects

Formulate a Dissertation Topic

- Refine and Approve
  - Draft a question: Turn your idea into one clear research question.
  - Ask for feedback: Share your draft title with an advisor or mentor to get official approval.

Research Proposal Components

- **Introduction**
- **Problem Statement**
- **Literature Review**
  - Related work
  - Comparison and contrast
  - Highlight the difference between your approach and others’
- **Research Methodology**
  - Data
  - methods
- **Expected Outcome**
  - Contribution & limitations
- **Expected Timeline**
- **Reference**